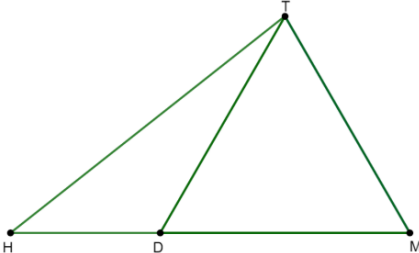
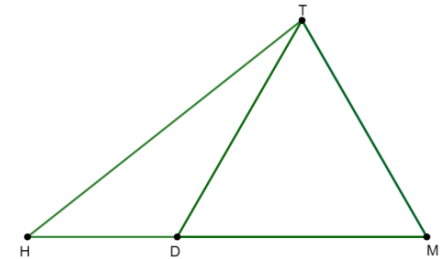


11.6 Triangle Inequality Theorem

Lesson Introduction

Lesson Introduction				
 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Targets	- I can determine if a set of triangle side lengths can be used to construct a triangle. - I can use the Triangle Inequality Theorem to prove relationships in triangles.
 Benchmark Coherence	Building On MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.		Working Toward N/A	
 Essential Questions	- How do you determine if a set of side lengths can be used to construct a triangle? - How can you use the Triangle Inequality Theorem to prove relationships in triangles?			
 Lesson Narrative	In this lesson, students discover how to use the Triangle Inequality Theorem to determine if a set of triangle side lengths can be used to construct a triangle. Students explore how to apply the theorem to prove that the order of the side lengths is the same as the order of their opposite angle measurements.			
 Component Summary	11.6.1 Warm-Up (~5 min.)	Solve a number puzzle (<i>SE p. 208</i>)	11.6.4 Exploration (5 min.)	Measure interior angles in triangle; list angles and side lengths from least to greatest (<i>SE p. 210</i>)
	11.6.2 Exploration (10 min.)	Explore if triangles can be constructed with side lengths (<i>SE p. 208</i>)	11.6.5 Guided Instruction (10 min.)	Discover how to prove inequality in a triangle (<i>SE p. 211</i>)
	11.6.3 Try It (5 min.)	Practice applying the Triangle Inequality Theorem (<i>SE p. 210</i>)	11.6.6 Try It (5 min.)	Order angles from least to greatest (<i>SE p. 212</i>)
	11.6.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 213</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Triangle Inequality Theorem - Whole Greater Than Parts Theorem <u>Additional Glossary:</u> - N/A		- Rulers - Protractors - Pieces of Paper	Prior to the lesson, work through 11.6.2 Exploration, 11.6.4 Exploration, and 11.6.5 Guided Instruction to ensure that you are familiar with the steps students will take and are prepared to support them as they work.

 Teacher Tips	Common Misconceptions - Students may not understand that the sum of any two sides in a triangle must be greater than the third side. - Students may not understand that the order of the measurements of the angles is the same as the order of their opposite side lengths. - Students may struggle with not having a visual when determining if given sides lengths create a triangle.	Things to Consider Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the side lengths in triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display During 11.6.6 Try It, consider using the instructional routine “Poll the Class” for question 1 for immediate feedback. Ask the students to hold up a whiteboard or piece of paper with the smallest angle written on it. Continue this for the next two angles. As students share their answer, scan the class to review students’ responses, and consider asking students to share why they chose that answer.	MA.K12.MTR.1.1: Participate in Effortful Learning 11.6.2 and 11.6.4 Explorations provide students with the opportunity to collaborate with their peers as they explore the relationships among the side lengths and angle measurements in triangles. These discussions will help students build confidence and understanding with the learning targets. Encourage students to justify their claims and discuss whether they agree or disagree with their peers and why.
 Differentiate Instruction	Consider using additional visuals, such as GeoGebra, to help students better conceptualize the concept of the Triangle Inequality Theorem for visual and kinesthetic learners. Students will have the chance to draw their own conclusions and discuss their rationale with peers, providing an opportunity for auditory learners to verbalize their understanding.	
		
Lesson Notes:		



11.6.1 Warm-Up: Number Riddle

Component Overview: Students solve a number puzzle by substituting operations in a number sentence to obtain a given value. Solving non-routine problems, such as this puzzle, can help students build strategic competence.



11.6.1 Warm-Up: Number Riddle

1. Use any of the four operations (+, −, ×, ÷) with the numbers shown to produce a result of 36.

$$7 \times 5 + 3 - 2 + 0$$

7, 0, 2, 3, 5



Encourage students to use the “guess, check, and revise” strategy to solve the puzzle. Remind students to keep track of their attempts, so that they do not repeat attempts that were incorrect. Prompt students to share what they learned from their mistakes, and how their mistakes helped them improve their guesses.

11.6.2 Exploration: Triangle Inequality Theorem

Component Overview: Students explore if a triangle can be constructed from given side lengths and create conjectures about how to determine if triangle side lengths can be used to construct a triangle.



11.6.2 Exploration: Triangle Inequality Theorem

1. With your partner, use a ruler to determine if the given measurements can be used to create a triangle. Then, complete the table.

Measurements in Centimeters (cm)	Can it create a triangle?	Explanation
3 cm, 6 cm, and 8 cm	☐ Yes ☐ No	<i>All the sides connect</i>
3 cm, 4 cm, and 10 cm	☐ Yes ☐ No	<i>All the sides don't connect</i>
8 cm, 11 cm, and 15 cm	☐ Yes ☐ No	<i>All the sides connect</i>
2 cm, 5 cm, and 8 cm	☐ Yes ☐ No	<i>All the sides don't connect</i>
4 cm, 5 cm, and 11 cm	☐ Yes ☐ No	<i>All the sides don't connect</i>

2. What similarities did you notice about the measurements that could create a triangle?
Answers will vary. When the sum of two sides is greater than the third, all the sides connect.
3. What similarities did you notice about the measurements that could not create a triangle?
Answers will vary. When the sum of two sides is less than the third, all the sides do not connect.
4. Make a conjecture about how to determine if side lengths can be used to construct a triangle if you cannot draw it.
Answers will vary. If the sum of any two side lengths is greater than the third, then a triangle can be constructed.



For this Exploration, allow students to engage with it with their partner as they create triangles using the given lengths. Provide students with rulers and paper to use throughout this Exploration. Observe and listen as students work with their partner and ask students to share their ideas with the class at the conclusion of this activity.



Collect and Display

Listen for, collect, and display student conjectures about the relationships among the side lengths in triangles. Then, engage students in a debrief discussion about the Triangle Inequality Theorem. Include the theorem on an anchor chart or Word Wall and encourage students to reference it in subsequent components.

11.6.2 Exploration: Triangle Inequality Theorem (continued)

5. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Conjecture	Initials
<i>Answers will vary</i>	
<i>Answers will vary</i>	

6. Review your conjecture with your partner and if necessary, make any changes.
7. Use your conjecture to determine if 3 cm, 4 cm, and 7 cm will create a triangle.
No because $3 + 4$ is not greater than 7.
8. Check your answer by constructing the triangle with a ruler.
9. What did you notice about the measurements 3 cm, 4 cm, and 7 cm?
Two of the sides together were the exact length of the third side.
10. Adjust your conjecture if needed.



Consider using a dynamic tool, such as GeoGebra, to provide entry points for visual and kinesthetic students to investigate the relationship between the side lengths in triangles.

Suggested GeoGebra Link:
<https://www.geogebra.org/m/K5CEeBEu>



Triangle Inequality Theorem

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.



If three segments have lengths such that the sum of any two is greater than the third, then a triangle can be constructed from the segments.

11.6.3 Try It: Triangle Inequality Theorem

Component Overview: Students practice applying the Triangle Inequality Theorem to determine if given side lengths can be used to construct a triangle.



11.6.3 Try It: Triangle Inequality Theorem

1. Complete the table.

Side Lengths	Work	Can it create a triangle?
57, 16, and 75	$57 + 16 < 75$ $57 + 75 > 16$ $75 + 16 > 57$	☐ Yes ☒ No
33, 12, and 25	$33 + 12 > 25$ $33 + 25 > 12$ $25 + 12 > 33$	☒ Yes ☐ No
21, 5, and 26	$5 + 21 = 26$ $5 + 26 > 21$ $26 + 21 > 5$	☐ Yes ☒ No
61, 100, and 61	$61 + 61 > 100$ $61 + 100 > 61$ $61 + 100 > 61$	☒ Yes ☐ No
48, 24, and 21	$21 + 24 < 48$ $21 + 48 > 24$ $48 + 24 > 21$	☐ Yes ☒ No



Provide a kinesthetic entry point by debriefing the activity with the following activity:

Designate one side of the room as “yes,” and the other side of the room as “no.” Read the side lengths aloud. Prompt students to walk to the side of the room that represents their answer. Ask students on each side of the room to justify their answers. Challenge students by including additional side lengths that they did not see in the component. The student with the most descriptive explanation of their answer could also be given the opportunity to create and call out the side lengths in the next round.



- How did you determine the answer?
- What mental math strategies did you use to calculate the sums?
- Did you test all of the possible pairs of sums of the side lengths?

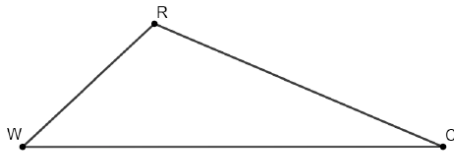
11.6.4 Exploration: Relating Angle Measures and Sides of Triangles

Component Overview: Students use protractors to measure the interior angles in a triangle. Then, they list the angle measurements and side lengths from least to greatest. Students discover that the order of the side lengths is the same as the order of their opposite angle measurements.



11.6.4 Exploration: Relating Angle Measures and Sides of Triangles

$\triangle RCW$ is shown, where $WC = 24$, $CR = 18$, and $RW = 10$.



1. Use a protractor to find the angle measures. Record your answers in the table.

Angle Measures
$m\angle CWR = 43^\circ$
$m\angle WRC = 114^\circ$
$m\angle RCW = 23^\circ$

2. List the side lengths in order from longest to shortest.
 WC, RC, WR
3. List the angles in order from largest to smallest.
 $\angle WRC, \angle CWR, \angle RCW$
4. What do you notice about the size of the angle compared to the length of the opposite side?
The longest side is opposite the larger angle and the shortest side is opposite the smaller angle.



Provide students the opportunity to create a conjecture about the size of the angle compared to the length of the opposite side. Allow students to share their conjecture with a partner to provide an entry point for auditory learners.

11.6.5 Guided Instruction: Proving Inequality in a Triangle

Component Overview: Students discover how to prove inequalities in a triangle. Engage students in discussions throughout the component to ensure that they understand the key terms and the proof.



11.6.5 Guided Instruction: Proving Inequality in a Triangle



If two **sides** of a triangle are unequal in measure, then the angle opposite the longer side has the greater measure.



If two **angles** of a triangle are unequal in measure, then the side opposite the angle with the greater measure is the longer side.

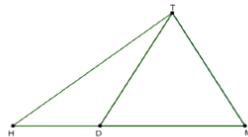


Whole Greater Than Parts Theorem

If $a > 0$, $b > 0$, and $a + b = c$, then $c > a$ and $c > b$.

$\triangle THM$ is shown, with $\overline{TD}$ creating $\triangle HTD$ and $\triangle DTM$, where $\triangle DTM$ is an equilateral triangle.

Given: $\triangle DTM$ is equilateral
Prove: $HM > HT$



Statement	Reason
1. $\triangle DTM$ is equilateral; $m\angle DTH = 22^\circ$	1. Given
2. $m\angle TMD = m\angle DTM$	2. Property of equilateral triangle
3. $m\angle DTM + m\angle DTH = m\angle MTH$	3. <i>Angle Addition Postulate</i>
4. $m\angle TMD + m\angle DTH = m\angle MTH$	4. Substitution Property of Equality
5. $m\angle MTH > m\angle TMD$	5. Whole greater than parts theorem
6. $HM > HT$	6. <i>If two angles of a triangle are unequal in measure, the side opposite the angle with the greater measure is the longer side.</i>



Add key terms and an additional glossary to the classroom Word Wall for students to reference in this and subsequent lessons.



- What do you know about the triangle based on the given information?
- How do the side lengths compare? How do the angle measurements compare?



Provide a visual entry point by prompting students to color code and label the diagrams as they work through each step of the proof.

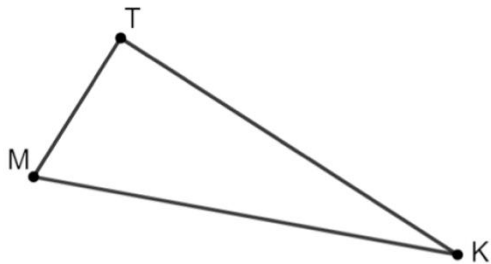
11.6.6 Try It: Triangle's Angle Measures

Component Overview: Students use the given triangle side lengths to order the angle measurements from least to greatest.



11.6.6 Try It: Triangle's Angle Measures

- Triangle TKM is shown, where $KT = 32$, $MK = 34$, and $TM = 13$. List the measures of the angles in order from least to greatest.



$m\angle TKM, m\angle TMK, m\angle MTK$



Poll the Class

Consider using the instructional routine “Poll the Class” for question 1 for immediate feedback. Ask the students to hold up a whiteboard or piece of paper with the smallest angle written on it. Continue this for the next two angles. As students share their answers, scan the class to review students’ responses, and consider asking students to share why they chose that answer.



What strategies did you use to help you determine your answer?

11.6.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.6.7 Wrap-Up: Cool Down

1. List three side lengths that would not make a triangle.
Explain how you know those sides would not make a triangle.

Answers will vary

2. List three side lengths that would make a triangle.
Explain how you know those sides would make a triangle.

Answers will vary